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Generalized Inverse

$A_{m \times n}$ has a right inverse if $\exists B_{n \times m}$ st

$$AB = I_m$$

$$\Updownarrow$$

$$\underbrace{\rho(A)}_{\text{Rank}} = m$$

$A_{m \times n}$ has a left inverse if $\exists C_{n \times m}$ st

$$CA = I_n$$

$$\Updownarrow$$

$$\rho(A) = n$$

If $b \in C(A)$,

~~$A_L^{-1}(A)$~~

$A_L^{-1} b$ & $A_R^{-1} b$ are Sols

to $AX = b$

For any g-inverse G,

$$A(Gb) = b$$

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Thm: Every matrix has a g-inverse.

Proof: If $A = O_{m \times n}$, inverse is $O_{n \times m}$.

Take RF (Rank Formation) of $A = (P, Q)$.

$A = P_{m \times r} Q_{r \times n}$ & $\rho(P) = \rho(Q) = r$

$G = (Q_R^{-1} P_L^{-1})_{n \times m}$

Take $b \in C(A)$

$AG(b) = A(Q_R^{-1} P_L^{-1}) Ax = PQQ_R^{-1} P_L^{-1} Ax$

$= Ax = b$

$\therefore$ G is a g-inverse of A but not all g-inverse of A are of this form.

Thm: $A_{m \times n}$ & $G_{n \times m}$

These are equivalent:

(i) G is a g-inverse of A.

(ii) $AGA = A$

(iii) AG is idempotent & $\rho(AG) = \rho(A)$

(iv) GA is idempotent & $\rho(GA) = \rho(A)$

Proof: (i) $\Rightarrow$ (ii)

$AGb = b$, $b = Ax$, $\forall x \in R^n$

$(AGA - A)x = 0$

$x \in N(AGA - A)$

$m \times n \quad n \times 1 \Rightarrow$ real

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Since, $\forall x$,

~~$AGA = AA_0$~~

$N(AGA - A) = \mathbb{R}^n$

OR

$\rho(AGA - A) = 0$

$\Downarrow$

$AGA = A$

(ii) $\Rightarrow$ (iii)

$AGA = A$

$AGAG = AG$

$(AG)^2 = (AG)$

$(AG)A = A$ ~~$AGA = A$~~ $\Rightarrow$ ~~$\rho(AG) \geq \rho(A)$~~ [G

$\rho(A) \geq \rho(AG)$ [AG is just A with operation. Can't create independent rows]

$\therefore \rho(AG) \geq \rho(A)$

(iii) $\Rightarrow$ (iv)

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(iv) $\Rightarrow$ (i)

$GAGA = GA$

$\rho(GA) = \rho(A)$

To show

$b \in C(A) \Rightarrow AG(b) = b$

Take, $b = Az, \quad z \in \mathbb{R}^n$

$\rho(GA) = \rho(A) \Rightarrow R(GA) = R(A)$

$\Downarrow$

$A = XGA$

$XGAGA = XGA$

$\Downarrow$

$AGA = A$

$$\boxed{AG(b) = AGAz = Az = b}$$

$Ax = b$ is consistent if $b \in C(A)$

$Ax = 0$ is always consistent [Set of soln of $N(A)$]

$Ax = b \Rightarrow$ Soln = Set of soln + $N(A)$

$A_{m \times n}$ is a matrix and G is a g-inverse of A.

(i) $Ax = b$ is consistent iff $AGb = b$

(ii) Solution space of $Ax = 0$ is $\{(I - GA)z : z \in \mathbb{R}^n\}$

(iii) If $Ax = b$ is consistent, soln is

$$\{Gb + (I - GA)z : z \in \mathbb{R}^n\}$$

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(i) $Ax = b$ is consistent $\Rightarrow b \in C(A)$, then by defn of g-inverse,

$AGb = b$

Take $x = Gb$.

$Ax = b \Rightarrow b \in C(A)$

(ii) To show $N(A) = \ell(I - GA)$

(column)

Take $x \in N(A)$.

$Ax = 0$

$AGAx = Ax$

$A(I - GA)x = 0$

$Ax = 0$

$GAx = 0$

$x = (I - GA)$

$\therefore x \in \ell(I - GA)$

$N(A) \subseteq \ell(I - GA)$

Take $x \in \ell(I - GA)$

$x = (I - GA)y$

$= y - GAy$

$Ax = Ay - AGAy$

$= Ay - Ay$

$= 0$

$\therefore x \in N(A)$

$\ell(I - GA) \subseteq N(A)$

(iii) G_b is a particular solution

↓

Done

A has a left inverse G_1 is a g-inverse & G_1 is a left inverse

$$AG_1A = A$$
$$A_L^T AG_1A = A_L^T A$$

$$G_1A = I$$

$\therefore G_1$ is Left inverse.

A has a right inverse $\Rightarrow$ G_1 is right inverse

Similar

If G_1 & G_2 are two inverses of A, $\alpha G_1 + (1-\alpha) G_2$ is an inverse.

$$A(\alpha G_1 + (1-\alpha) G_2)A = \alpha A G_1 A + (1-\alpha) A G_2 A$$

$$= \alpha A + (1-\alpha) A$$

$$= A$$

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$A_{m\times n}$, $G_{n\times m}$

$$\rho(A) \le \rho(G) \le \min\{m,n\}$$

$$AGA = A$$

$$\rho(A) \le \rho(AG) \le \rho(G)$$

$A = PQ$ is an RF of A,
then $G = Q_R^{-1} P_L^{-1}$ is g inv of A

Then $\rho(G) = \rho(A)$

Thm → For any s st $\rho(A) \le s \le \min\{m,n\}$, $\exists$ g inv of A, say G, with $\rho(G) = s$

Proof →

$$E = \begin{bmatrix} X_{r\times r} & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} \quad \text{st } X^{-1} \text{ exists}$$

$$G = \begin{bmatrix} X^{-1} & B \\ C & D \end{bmatrix}_{n\times m} \quad \text{is g inv of } E$$

$$EGE = \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X^{-1} & B \\ C & D \end{bmatrix} \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} I_r & XB \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix} = E$$

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Suppose $A_{m\times n}$ is a matrix with $\rho(A) = r$,

Then, $A = P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} Q$ where (Row Echelon form)

$P_{m\times m}$ and $Q_{n\times n}$ are non-singular matrices.

$A_{m\times n} = \bar{P}_{m\times r}\, \bar{Q}_{r\times n}$, where $\rho(A) = r$

Construct $P = [\bar{P}\ \hat{P}]_{m\times m}$ where columns of P form a basis of $\mathbb{R}^m$. This means P is a non singular matrix. [Add $\hat{P}$ which have more independent vectors]

Similarly, $Q = \begin{bmatrix} \bar{Q} \\ \hat{Q} \end{bmatrix}_{n\times n}$, where Q forms basis of $\mathbb{R}^n$. $\therefore Q$ is non singular.

$$A = [\bar{P}\ \hat{P}] \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \bar{Q} \\ \hat{Q} \end{bmatrix}$$

$$= [\bar{P}\ 0] \begin{bmatrix} \bar{Q} \\ \hat{Q} \end{bmatrix}$$

$$A = \bar{P}\bar{Q}$$

$$\therefore A = P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q,$$

where P, Q are non singular.

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$Q^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}^{-} P^{-1}$ is a g-inverse of A.

$\Rightarrow Q^{-1}\begin{bmatrix} I_r & B \\ C & D \end{bmatrix} P^{-1}$ is a g-inverse of A

Take $B = C = 0$, and D has first $s-r$ rows independent.

Result → $\rho\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\right) \geq \rho(A) + \rho(D)$ (Verify)

Using this, we can show that $\rho\left(\begin{bmatrix} I_r & B \\ C & D \end{bmatrix}\right) \geq r+s-r$

$\geq s$

Since only s non-zero rows, $\rho\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\right) \leq s$

$\therefore \rho(\text{g-inv}) = s$

Computation of g-inverse

$E_{m\times m} A_{m\times n} = F_{m\times n} \rightarrow$ $p_1 < p_2 < \ldots$

$$\begin{bmatrix} 1 & 0 & 0 & \star & \cdots \\ 0 & 0 & 1 & \cdots & \\ \cdots & & & & \end{bmatrix}$$

G_1 is 1st row of

Construct $G_{n\times m}$ where p_1th row of G is 1st row of E, ..., p_rth row of G is rth row of E.

To show that G is g-inverse of A.

$\rho(GA) = \rho(A)$ and GA is idempotent.

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P_ith row of GA will be ith row of $EA = F$, and other rows are 0.

~~$\therefore \rho(F) = \rho(A)$~~

$\therefore \rho(GA) = \rho(EA)$

$\rho(EA) = \rho(A)$

$\Downarrow$

$\boxed{\rho(GA) = \rho(A)}$

GA has r non null rows.

$P_1, P_2, \ldots P_r$ rows are non null.

$GA \rightarrow$ 1st - (P_1-1) th column $\rightarrow$ 0

P_1th column $\rightarrow e_{P_1}$

(P_1+1)th column - (P_2-1)th column $\rightarrow \alpha_j e_{P_1}$

$[j \in [P_1+1, P_2-1]]$

P_2th column $\rightarrow e_{P_2}$

(P_2+1)th column - (P_3-1)th column $\rightarrow \beta_j e_{P_1} + \gamma_j e_{P_2}$

$\vdots$

$(GA)(GA) = GA$

$\Downarrow$

$(GA)(GA)_j = (GA)_j \quad \forall 1 \le j \le n$

$\Downarrow$

Till (P_1-1), each 0

$P_1 \rightarrow (GA)e_{P_1} = (GA)_{P_1} = e_{P_1}$

And it works.

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$\therefore$ G is g-inverse

Lemma - Amxn. Then, $A^T(AA^T)^{-1}$ and $(A^TA)^{-1}A^T$ are g-inv of A.

Proof - $AA^T(AA^T)^{-1}AA^T = AA^T$ $[AGA = A]$

$$\rho(AA^T) = \rho(A)$$
$$\Leftrightarrow C(AA^T) = C(A)$$

$$A = AA^TC$$

$$AA^T(AA^T)^{-1}AA^TC = AA^TC$$
$$AA^T(AA^T)^{-1}A = A$$

Some Properties of G

1) Reflexivity : $\rho(G) = \rho(A) \Leftrightarrow$ ~~$\rho(GAG) = \rho$~~ $GAG = G$

2) Minimum Norm : $b \in C(A) \Rightarrow \|Gb\| \leq \|x\|$, $\forall x$ st $Ax = b$
$\Leftrightarrow (GA)^T = GA$

3) Least Square : $\forall x \in \mathbb{R}^n, \forall y \in \mathbb{R}^m$
$\|AGy - y\| \leq \|Ax - y\|$
$\Leftrightarrow (AG)^T = AG$

1 $\Rightarrow$ $\rho(A) = \rho(AG) \leq \rho(G) \Rightarrow \rho(G) = \rho(AG) \Rightarrow G = CAG$

$AGA = A$

$GAG = CAGAG$

$GAG = CA$

$AGA = ACA$

$GAG = CAG$

$\boxed{GAG = G}$

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$\Leftarrow$

$GAG = G \Rightarrow \quad \rho(G) \leq \rho(AG) = \rho(A)$

$\rho(G) \leq \rho(A)$

$\rho(G) \geq \rho(A)$ [Basic prop]

$\therefore \boxed{\rho(A) = \rho(G)}$

2) $\Leftarrow$

$b \in \mathcal{C}(A) \Rightarrow Ax = b \Leftrightarrow x = Gb + (I - GA)z, \quad z \in \mathbb{R}^n$

$\|x\|^2 = \|Gb\|^2 + \|(I-GA)z\|^2 + 2b^T G^T (I-GA)z$

$b = Aw$

$2b^T G^T (I - GA)z$

$= 2w^T A^T G^T (I - GA)z$

$= 2w^T (GA - GAGA)z$

$= 2w^T (GA - GA)z$

$= 0$

$\therefore \|x\| \geq \|Gb\|$

$\Rightarrow$

$\|Gb\|^2 + \|(I-GA)z\|^2 + 2b^T G^T (I-GA)z \geq \|Gb\|^2$

$\|(I-GA)z\|^2 + 2b^T G^T (I-GA)z \geq 0$

$\|\ \|^2 + 2w^T A^T G^T (I-GA)z \geq 0$

Replace w with αw.

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$2\alpha w^T A^T G^T (I - GA) z$ ...

If ... $w^T A^T G$ ... is negative, we can choose α large enough such that the expression < 0.

If ... is positive, we can choose α to be small enough such that exp < 0.

$\therefore \quad w^T A^T G^T (I - GA) z = 0 \quad \forall w, \forall z$

$\therefore \quad A^T G^T (I - GA) = 0$

$GA^T = GA^T GA$

$GA = (GA)^T GA$ [Transpose both sides]

$\therefore \quad \boxed{GA = GA^T}$

3) $\Leftarrow$

$\|Ax - y\|^2 = \|AGy - y + A(x - Gy)\| = \|AGy - y\|^2 +$ ~~$\|Ax - AGy\|$~~ $\|Aw\|^2 +$ $2w^T A^T (AGy - y)$

$2w^T A^T (AGy - \frac{I}{y}) y = 2A^T 2w^T (A^T AG - A^T) y$

$= 2w^T (A^T (AG)^T - A^T) y$

$GAA^T - A^T$

$= 0$

$\therefore \quad \boxed{\|Ax - y\| \geq \|AGy - y\|^2}$

Answer Set 1

1 GA is idempotent

$\Downarrow$

$\rho(GA) = tr(GA)$

$B_{m\times n} = P_{m\times r} Q_{r\times n}$

$B^2 = B$

$PQPQ = PQ$

$P_L^{-1} P Q P Q Q_R^{-1} = P_L^{-1} P Q Q_R^{-1}$

$QP = I_r$

$tr(QP) = tr(PQ) = r$

$= \rho(B)$

$\rho(GA) = \rho(A)$

$\therefore$ $\boxed{\rho(A) = tr(GA)}$

2 $AGA = A$

$A - AGA = 0$

$A(I - GA)x = 0, \quad \forall x \in \mathbb{R}^n$

Take $y \in \mathcal{C}(I - GA) \Rightarrow y = (I - GA)x$

$Ay = 0$

$\Downarrow$

$y \in \mathcal{N}(A)$

$GAy = 0$

$y - GAy = y$

$(I - GA)y = y$

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$y \in \mathcal{C}(I - GA)$

$\rho(I - GA) = n - \rho(A)$

$d(\mathcal{C}(I-GA)) = d(N(A))$ [Rank-Nullity]

$N(A) \subseteq \mathcal{C}(I-GA)$ [Easy, Done]

$\therefore N(A) = \mathcal{C}(I-GA)$

Take $y \in \mathbb{R}^n$

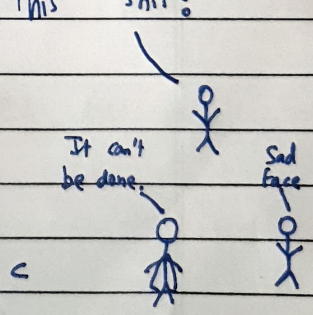
$(I-GA)y \in \mathcal{C}(I-GA)$

$A(I-GA)y = 0$

$A = AGA$

3. $\rho(C) \leq \rho(A)$

$\therefore C = AB$



$AGC = AGAB = AB = C$

$R(C) \subseteq R(A)$

$C = BA$

$CGA = BAGA = \cancel{AGA = A}\ BA = C$

$C_i(GA) = C_i, \forall i$

$C_i \in \mathcal{C}(A)$

$AGC_i = C_i$

$AGC = C$

G is a generator of A+B

$$P(A+B) = P(A) + P(B) \Rightarrow \{1\} \cap \{1\} = \{0\}$$

$$\{(A+B) = \{(A) + \{(B)$$

$$\{(A) \leq \{(A+B)$$

$$(A+B)\,GA_j = A_j \quad \forall j$$

$$\underbrace{AGA_j}_{\{(A)} + \underbrace{BGA_j}_{\{(B)} = A_j$$

$$AGA_j = A_j$$
$$BGA_j = 0$$

$\rho(AB) = \rho(B) - \dim\left(\mathcal{C}(B) \cap \mathcal{N}(A)\right)$

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3) Least Square cont

$\Rightarrow$

$\|AGy - y\| \leq \|Ax - y\| \quad , \forall x, y$

$\|AGy - y\| \leq \|AGy - y + Aw\| \quad , w = x - Gy$

$\|AGy - y\| \leq \|AGy - y\|^2 + \|Aw\|^2 + 2w^T A^T (AGy - Iy)$

$\|Aw\|^2 + 2w^T A^T (AG - I)y \geq 0$

↓

We can always choose y to screw things up, so

$2w^T A^T (AG - I)y = 0$

$w^T A^T (AG - I) = 0$

$G^T A^T (AG - I) = 0$

$(AG)^T (AG) A = (AG)^T$

$(AG)^T (AG) = AG$

$\therefore AG = AG (AG)^T$

Moore - Penrose g-inverse

(i) Reflexive (ii) Minimum Norm (iii) Least Square

⇓

(i) $AGA = A$ (ii) $GAG = G$ (iii) $GA^T = GA$ (iv) $(AG)^T = AG$

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Thm → Every matrix has a MP g-inverse.

Proof → If $A_{m\times n} = 0$, $G_{n\times m} = 0$ is the MP g-inverse.

Suppose A is non null.

$$A = P_{m\times r}\, Q_{r\times n}$$

$$G = Q^T (P^T A Q^T)^{-1} P^T$$

$\rho(G) \le \rho(Q^T) = r$
$\rho(G) \ge \rho(A) = r$
$\therefore \rho(G) = r$

$$AGA = PQ\,Q^T (P^T PQ\,Q^T)^{-1} P^T PQ$$
$$= PQ$$
$$AGA = A$$

$$GAG = Q^T (QQ^T)^{-1} (P^T P)^{-1} P^T PQ\, Q^T (QQ^T)^{-1} (P^T P)^{-1} P^T$$
$$= Q^{-1} P^{-1}$$
$$= G$$

$$GA = Q^{-1} P^{-1} PQ$$
$$= I$$

$$GA = Q^T (P^T A Q^T)^{-1} P^T PQ$$
$$(GA)^T = Q^T P^T P (QA^T P)^{-1} Q$$
$$= Q^T P^T P P^{-1} (A^T)^{-1} Q^{-1} Q$$
$$GA^T = Q^T P^T (A^T)^{-1}$$

∎

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Thm → MP g-inverse is unique.

Proof → Take two MP g-inverses, say G_1 and G_2.

$$G_1 = G_1AG_1 = G_1G_1^TA^T = G_1G_1^TA^TG_2^TA^T$$
$$= G_1AG_1G_2^TA^T$$
$$G_1 = G_1AG_2$$

$$G_2 = G_2AG_2 = A^TG_2^TG_2 = A^TG_1^TA^TG_2^TG_2 = A^TG_2^TG_2AG_2$$
$$= G_1AG_2$$

$$\therefore \boxed{G_1 = G_2}$$

Projection

S and T are subspaces of $\mathbb{R}^n$.

$$S + T = \{u \in \mathbb{R}^n \mid u = x+y, \ x \in S \text{ and } y \in T\}$$

$$S \oplus T = \{u \in \mathbb{R}^n \mid u = x+y, \ x \in S \text{ and } y \in T, \text{ and } x, y \text{ are unique}\}$$

A subspace T of $\mathbb{R}^n$ is a complement of S (subspace of $\mathbb{R}^n$) if

$$S \oplus T = \mathbb{R}^n$$
and $S \cap T = \emptyset$ $\{0\}$

Defn → Let T be a complement of a subspace S and $u \in \mathbb{R}^n$. Then projection of u into S along T is x, where $u = x+y$, where $x \in S$ and $y \in T$

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Thm → Let T be a complement of S and $P(u)$ denote the projection of u into S along T.
Then,

(i) Projection into T along S is $u - P(u)$

(ii) $P(u) = u$ iff $u \in S$

(iii) $P(u) = 0$ iff $u \in T$

(iv) $P(P(u)) = P(u)$

(v) $P(u+v) = P(u) + P(v)$

(vi) $P(\alpha u) = \alpha P(u)$

Proof → (i) $u = P(u) + y$

$y = u - P(u)$

(ii) $y = u - u$

$y = 0$

(iii) $y = u - 0$

$y = u$

(iv) $P(u) \in S$

$P(P(u)) = P(u)$

(v) $u = x + y$ ($x \in S$)

$v = w + z$ ($w \in S$)

$P(u) + P(v) = x + w$

$P(u+v) = x + w$ $[x+w \in S,\ y+z \in T]$

(vi) $u = x + y$

$\alpha u = \alpha x + \alpha y$

$\alpha x \in S$

$\alpha y \in T$

$P(\alpha u) = \alpha P(u)$

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$S, T \subseteq \mathbb{R}^n$

S and T are complement of each other. We will try to get a matrix $P_{n\times n}$ st for any vector $u \in \mathbb{R}^n$, Pu is projection of u into S along T.

Defn: A matrix $P_{n\times n}$ is called a projector into S along T if for $u \in \mathbb{R}^n$, Pu is the projection of u into S along T.

Thm: The following are equivalent :-

(i) P is a projection matrix

(ii) $P^2 = P$

(iii) $N(P) = C(I-P)$

(iv) $\rho(P) + \rho(I-P) = n$

Proof: (i) $\Rightarrow$ (ii)

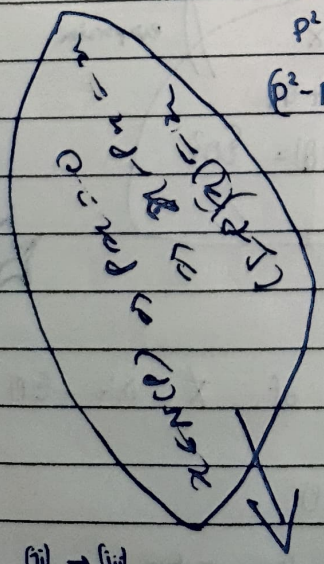
Assume P projects into S along T. Take $x \in \mathbb{R}^n$

$Px \in S$

$P(Px) = Px$

$P^2x = Px$

$(P^2-P)x = 0, \quad \forall x \in \mathbb{R}^n$



$x \in N(P^2-P)$

$N(P^2-P) = \mathbb{R}^n$

$\rho(P^2-P) = n-n = 0$

$\therefore P^2 = P$

(ii) $\Rightarrow$ (iii)

$x \in N(P) \quad N(P) \subseteq C(I-P)$ (General)

$Px = 0$ Take $x \in C(I-P)$

$P^2x = 0$ $x = (I-P)z$

$Px = (P-P^2)z = 0 \quad \therefore x \in N(P)$

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(iii) $\Rightarrow$ (iv)

$$d(N(P)) = d(\mathcal{C}(I-P))$$

$$n - \rho(P) = \rho(I-P)$$

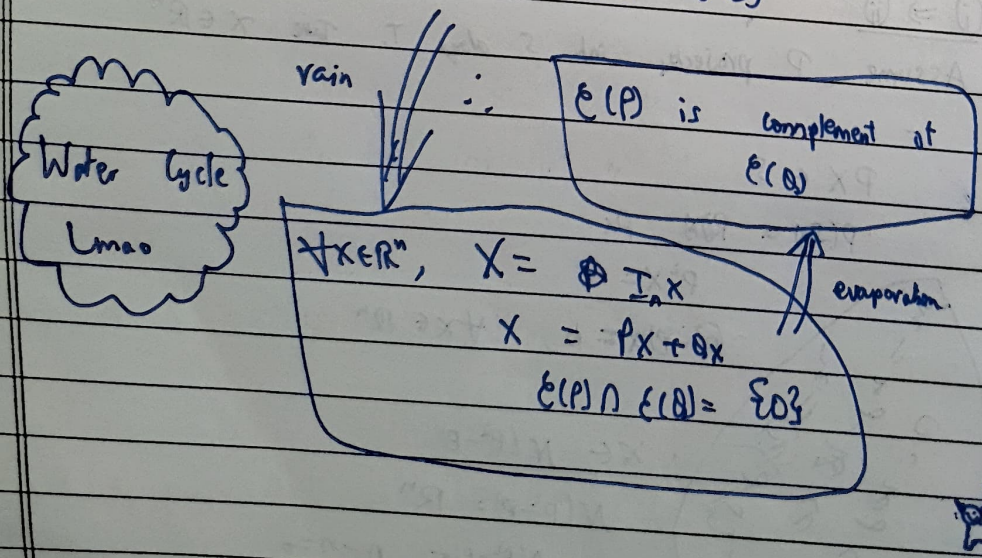
$$\rho(P) + \rho(I-P) = n$$

(iv) $\Rightarrow$ (i)

$$I - P = Q$$

$$P + Q = I$$

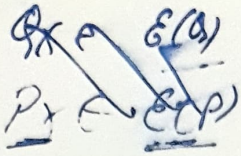
$$\rho(P) + \rho(Q) = \rho(I)$$





$\therefore$ Px is projection of x into $\mathcal{C}(P)$ along $\mathcal{C}(Q)$.
$\Rightarrow$ $\mathcal{C}(I-P)$

$\therefore$ P is projection matrix.



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If T is complement of S and P is projection into S along T. P is unique.

↓

Projection of e_1 is unique $\leftarrow Pe_1 = P_1 \rightarrow$ ∴ First column is unique, and so on...

$Pe_2 = P_2$

⋮

↓

∴ P is unique.

Projection Matrix exists for all S, T.

Construction

Let T be a complement of S. Take a basis of S and a basis of T. Construct

$$A = \underset{n\times r}{[B} \; \underset{n\times n-r}{C]}, \quad \mathcal{E}(B) = S, \quad \mathcal{E}(C) = T$$

$$\mathcal{V}(A) = n$$

Write $A^{-1} = \begin{bmatrix} G \\ \hline H \end{bmatrix}$ st $\underset{n\times n}{BG}$ is defined.

Claim: BG is a projector into S along T.

Proof: $AA^{-1} = I \Rightarrow [B|C]\begin{bmatrix} G \\ \hline H \end{bmatrix} = I$

$$BG + CH = I$$

$$\forall x \in \mathbb{R}^n, \quad x = \underbrace{BGx}_{S} + \underbrace{CHx}_{T}$$

$\mathcal{E}(BG) \subseteq \mathcal{E}(B)$,

$\mathcal{E}(CH) \subseteq \mathcal{E}(C)$

$\Downarrow$

$\mathcal{E}(BG) \cap \mathcal{E}(CH) = \{0\}$

$\mathcal{E}(BG) \subseteq \mathcal{E}(B)$

$\rho(BG) \leq r$

$\rho(BG) \geq \rho(B) + \rho(G) - r$

$\geq r + r - r$

$\geq r$

$\rho(BG) = \rho(B)$

$\rho(CH) = \rho(C)$

S is a subspace of $\mathbb{R}^n$, and $x \in \mathbb{R}^n$.

Find $\min_{w \in S} \|x - w\|$.

$S^{\perp} = \{x \in \mathbb{R}^n : \langle x, y \rangle = 0, \forall y \in S\}$

w is the orthogonal projection of x into S (along $S^{\perp}$)

$\Downarrow$

Orthogonal Projection

$\|x - w\|^2 = \|x - op(x) + op(x) - w\|^2$

$= \|x - op(x)\|^2 + \|op(x) - w\|^2 + 2\,(op(x) - w)^T (x - op(x))$

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$X - op(x) \in S^{\perp}$

$w - op(x) \in S$

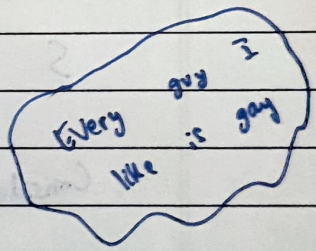
$\therefore \quad \|X - w\|^2 = \|X - op(x)\|^2 + \|op(x) - w\|^2$

$\geq \|X - op(x)\|^2$

$\therefore \quad \boxed{w = op(x)}$

Orthogonal Projection Matrix

P is orthogonal projection matrix $\iff$ P is idempotent and symmetric



$P = P^2$

$= PP$

$P = P^T P$

Since $P = P^2$,

P projects into $\mathcal{C}(P)$ along $\mathcal{C}(I-P)$

To show $\mathcal{C}(P) \perp \mathcal{C}(I-P)$

Take vectors Px and $(I-P)y$

$\langle Px, (I-P)y \rangle = x^T P^T (I-P) y$

$= x^T (P^T - P^T P) y$

$= x^T (P - P^2) y$

$= 0$

Woo hoo

$\therefore \quad \mathcal{C}(P) \perp \mathcal{C}(I-P)$

Converse

Suppose, P is orthogonal projection matrix.

$P^2 = P$ & $\mathcal{C}(P) \perp \mathcal{C}(I-P)$

$x^T P^T (I-P) y = 0$

$x^T (P^T - P^T P) y = 0$

$\forall x, y$

$\therefore \quad P^T = P^T P$

$P = P^T P$

$\therefore \quad \boxed{P = P^T}$

Computation of orthogonal projection matrix

S

Consider $A_{m \times n}$ st $\mathcal{C}(A) = S$

$P_A = A (A^T A)^- A^T$

is projection matrix ~~is S.~~ into S.

Need to show : (i) $P_A = P_A^T P_A$

(ii) $\mathcal{C}(P_A) = \mathcal{C}(A)$

(i) $A (A^T A)^- A^T = A \underbrace{(A^T A)^- A^T}_{\text{g-inverse of } A.} A (A^T A)^- A^T$

$= A (A^T A)^- A^T$

(ii) $\mathcal{C}(P_A) \subseteq \mathcal{C}(A)$

$P_A A = A$

$\therefore \quad \mathcal{C}(A) \subseteq \mathcal{C}(P_A)$

$\therefore \quad \mathcal{C}(P_A) = \mathcal{C}(A)$

$\therefore$ P_A is orthogonal projection matrix into S.

$\mathcal{C}(A) = \mathcal{C}(B) \Rightarrow P_A = P_B$

$\mathcal{C}(A) = \mathcal{C}(D) \Rightarrow P_A = P_D$

D is matrix with orthonormal columns.

$P_D = D(D^TD)^{-1}D^T$

$= DD^T \quad [D^TD = I_r]$

$S_1 + S_2 + \ldots + S_k$

$S_1 + S_2 + S_3 = Sp(S_1 \cup S_2 \cup S_3)$

$\Downarrow$

ppt

$Sp(S_1 \cup S_2 \cup S_3) \subseteq S_1 + S_2 + S_3$

↳ Sum of subspaces is also a subspace.

$S_1 \subseteq S_1 + S_2 + S_3$

$\Downarrow$

$US_1 \subseteq S_1 + S_2 + S_3$

$\Downarrow$

$Sp(US_1) \subseteq S_1 + S_2 + S_3$

$S_1 + S_2 + S_3 \subseteq Sp(S_1 \cup S_2 \cup S_3)$

$x \in S_1 + S_2 + S_3$

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$$X = \underset{\in S_1}{x_1} + \underset{\in S_2}{x_2} + \underset{\in S_3}{x_3}$$

$x_1 \in sp(S_1 \cup S_2 \cup S_3)$

$x_2 \in sp(S_1 \cup S_2 \cup S_3)$

$x_3 \in sp(S_1 \cup S_2 \cup S_3)$

$\therefore x \in sp(S_1 \cup S_2 \cup S_3)$

$\therefore$ $\boxed{S_1 + S_2 + S_3 = sp(\cup S_i)}$

$S_1 \oplus S_2 \oplus S_3$ ✗ $\quad S_i \cap S_j = \{0\}, \ \forall i,j \in \{1,2,3\}$

$\Downarrow$

$\{(1,0,0)\}, \{(0,1,0)\}, \{(1,1,0), (0,0,1)\}$

$\Leftrightarrow \quad S_3 \cap S_2 = \{0\}$ & $(S_1 + S_2) \cap S_3 = \{0\}$

Thm - Equi →

① $S_1 + S_2 + S_3$ is direct.

② $S_1 \cap S_2 = \{0\}$ and $(S_1 + S_2) \cap S_3 = \{0\}$

③ $d(S_1 + S_2 + S_3) = d(S_1) + d(S_2) + d(S_3)$

Proof - ① → ②

$x \in S_1 \cap S_2$

$\Rightarrow \quad x \in S_1 + S_2 + S_3$

$x = \underset{\in S_1}{x} + 0 + 0 = 0 + \underset{\in S_2}{x} + 0$

Since decomposition is unique,

$x = 0$

$\therefore S_1 \cap S_2 = \{0\}$

$x \in (S_1 + S_2) \cap S_3$

$x = x_1 + x_2 + 0 = 0 + 0 + x$

$\Downarrow$

$\boxed{x = 0}$

(ii) $\Rightarrow$ (i)

Assume sum is not direct.

$x \in S_1 + S_2 + S_3$

$x = y_1 + y_2 + y_3 = z_1 + z_2 + z_3$.

$y_i \neq z_i$

$y_3 - z_3 = (z_1 - y_1) + (z_2 - y_2)$

$y_3 - z_3 \in S_3$, $z_1 - y_1 \in S_1$, $z_2 - y_2 \in S_2$

But $S_3 \cap (S_1 + S_2) = \{0\}$

Contra.

$\therefore$ Sum is direct.

(iii) $\Rightarrow$ (ii)

$d(S+T) = d(S) + d(T) - d(S \cap T)$

$d(S_1 + S_2 + S_3) = d(S_1 + S_2) + d(S_3)$

$d(S_1 + S_2 + S_3) = d(S_1) + d(S_2) + d(S_3)$

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(iii) ⇒ (ii)

$$d(S_1 \oplus S_2 \oplus S_3) \leq d(S_1 \oplus S_2) + d(S_3)$$

$$\Downarrow$$

$$(S_1 \oplus S_2) \cap S_3 = \{0\}$$

$$d(S_1 + S_2) \leq d(S_1) + d(S_2)$$

$$\Downarrow$$

$$S_1 \cap S_2 = \{0\}$$